

Synergistic Activities

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1. Open Robotics Knowledge Dissemination

Created and publicly shared technical portfolios that translate graduate-level robotics work into accessible engineering resources covering perception, behavioral cloning, learned forward dynamics, model predictive control, and reinforcement learning. The repositories document system architectures, methodology, validation results, and reusable interfaces to support students and early-career engineers.

2. Hardware-Software Integration and Safety Knowledge Transfer

Documented the development of a Jetson-based CAN-bus robotic neck control system integrating RMD-X6 motors, inverse kinematics, dual-motor synchronization, and motion smoothing. The project also communicates practical lessons from mechanical instability, material failure, electrical short-circuit risk, and physical prototype validation, helping others recognize integration and safety constraints in robotic systems.

3. Multidisciplinary Rapid Prototyping and Student Innovation

Served as team lead for a real-time face-tracking robotic-eye prototype developed during Columbia University Robotics Club's Hack & Build hardware hackathon. The system integrated computer vision, embedded motor and LED control, custom CAD, and physical prototyping; it received third place and demonstrated how a multidisciplinary team can rapidly translate a concept into a functional robotic system.

4. Translation of AI and Data Methods into Manufacturing Practice

Applied computer vision, data analysis, design of experiments, and process automation to optoelectronic manufacturing challenges, including appearance inspection, defect identification, equipment validation, and production workflow improvement. These efforts transferred analytical and AI methods into practical industrial tools that improved inspection efficiency, defect reduction, and manufacturing readiness.

5. Engineering Training and Workforce Development

Developed standard operating procedures, equipment-validation methods, and structured technical training for manufacturing technicians and cross-functional engineering teams. These activities supported adoption of new inspection systems, automation workflows, and production processes while improving consistency, technical understanding, and operational performance.